

TUNIR SIKDAR

Data Engineer II | Deloitte USI

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PROFESSIONAL SUMMARY

Data Engineer with **5 years** of experience architecting and scaling **distributed data platforms** on **AWS and Azure**. Skilled at translating ambiguous data problems, missing feeds, inconsistent vendor formats, historical gaps, into **fault-tolerant, idempotent pipeline designs** and **Lakehouse architectures** that support high-volume, mission-critical workloads. Owns system design end-to-end, from ingestion and orchestration to storage and governance, improving reliability, performance, and cost efficiency for Fortune 500 clients across insurance, retail, and e-commerce.

SKILLS

- **Languages:** Python, SQL, T-SQL
- **System Design:** Distributed systems architecture, fault-tolerant & idempotent pipeline design, Lakehouse & medallion architecture, SLA-driven orchestration, scalability planning
- **AWS:** Lambda, Glue, S3, EC2, CloudWatch, Secrets Manager
- **Azure:** Databricks, Data Factory (ADF), Data Lake Storage (ADLS Gen2), Key Vault, Azure SQL Database
- **Big Data:** Apache Spark, Apache Iceberg, Hadoop (basic)
- **Data Engineering:** Data modeling & normalization, data quality tooling (Great Expectations), imputation & backfill strategies
- **DevOps & Tools:** Azure DevOps, Git, CI/CD

EXPERIENCE

Data Engineer II, Deloitte USI – Kolkata, India

04/2025 – Present

- Solved the challenge of onboarding **17 disparate vendor feeds** into a single insurance data platform by architecting a config-driven, idempotent ingestion framework, including **backfill logic** to safely reprocess historical batch data for newly onboarded sources, scaling to **4.2 TB of daily batch volume** with no duplication or data loss on reprocessing.
- Designed **imputation and standardization rules** to handle missing or malformed fields across third-party feeds, and defined the **Lakehouse table layout, partitioning, and compaction strategy** that kept query performance stable as data volume scaled.
- Architected a **fault-tolerant orchestration layer** on **AWS Lambda** and SQL Server control tables with automatic retry and checkpointing, sustaining a **99%+ on-time SLA** for commission-critical reporting.

Firm Initiatives

- Led a team of **5 data engineers** to build a scalable, **real-time event ingestion and analytics pipeline** for a mobile application using **Spark Structured Streaming**, powering operational dashboards used by business stakeholders to monitor live usage metrics.
- Architected a **Lakehouse platform** for a fast-growing e-commerce client using **Databricks Autoloader** for incremental ingestion, scaling to support **3x year-over-year growth** in transaction volume across curated reporting layers.

Data Engineer, Fractal Analytics – Pune, India

09/2023 – 03/2025

- Delivered **24 end-to-end big data projects** on Azure, designing scalable **PySpark and Spark SQL** jobs on datasets exceeding **500+ GB** that improved execution efficiency by **over 40%**.
- Solved recurring data gaps in a **Fortune 500 retail client's** inventory feeds by designing **config-driven ETL pipelines** with **imputation logic** for missing SKU-level fields and automated **backfill jobs** to reprocess historical records, structuring output into **Delta tables** with star and snowflake schema modeling for reliable BI reporting at scale.
- Reduced manual intervention in error handling by **60%** by designing automated data validation and monitoring frameworks in **PySpark** and cut annual licensing costs by **\$150K** by re-architecting workflows onto Azure.

Application Developer, IBM India Pvt. Ltd. – Kolkata, India

02/2021 – 08/2023

- Migrated **40+ business-critical pipelines** from legacy systems to Azure using **ADF** and built data quality validation checks in Databricks and Python.
- Reduced manual data preparation time by **42 hours per month** by streamlining extraction processes and cutting operating costs by **30%** through faster issue resolution.

EDUCATION

B.Tech. in Electronics and Communication Engineering, Heritage Institute of Technology

08/2016 – 08/2020

GPA: 9.09/10.0

CERTIFICATIONS

- Databricks Certified Data Engineer Associate (2025)
- Fractal Certified PySpark Developer

AWARDS

- Outstanding Award, Deloitte USI
- Applause Award, Deloitte USI
- Star Award (Kaizen), Fractal Analytics
- Manager Appreciation Award, IBM India Pvt. Ltd.